

MD MAZHARUL ISLAM EMON

AI-Native Software Engineer

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Date of birth: 26.07.1999 • Nationality: Bangladeshi

PROFILE

Software engineer with 3+ years building AI-powered, production-scale SaaS platforms used by tens of thousands of test-prep candidates across IELTS and PTE. Architected multi-tenant Django/DRF backends, real-time speech-evaluation pipelines, and Flutter/Next.js clients; shipped 90+ production releases across iOS, Android, and Web. AI-native engineer who ships faster using Claude Code and agentic workflows for code generation, refactoring, and test scaffolding. Competitive programming background: 2,000+ problems solved, ICPC Asia-Dhaka Regional finalist.

EDUCATION

Leading University Sylhet, Bangladesh
B.Sc. in Computer Science & Engineering — GPA: 3.6 / 4.0 09/2018 – 12/2022

- Relevant coursework:** Data Structures, Algorithms, Operating Systems, Computer Networks, Databases, Distributed Systems, Machine Learning.
- Final-year project — Projecto:** multi-role university course & proposal management system built with Django, with role-based workflows, automated notifications, and PDF/CSV reporting.
- Leadership:** Founding Chair, IEEE Computer Society LU Student Branch (2022–23); ACM Coordinator, LU Computer Club (2022–23).

LANGUAGES

Bengali: Native
English: Full professional — IELTS Academic 6.5 (Listening 7.0 · Reading 6.5 · Writing 6.5 · Speaking 6.0), Aug 2025 · CEFR B2

TECHNICAL SKILLS

Languages: Python · TypeScript · Dart · SQL · JavaScript · C++
Backend & APIs: Django · Django REST Framework · FastAPI · REST · WebSockets · Celery · async Python
Frontend & Mobile: Flutter · Riverpod · Next.js · React · Tailwind CSS · Zod
Data & Infra: PostgreSQL · Redis · Docker · Nginx · GCP · DigitalOcean · CI/CD · Prometheus · Grafana
AI / ML: OpenAI API · WhisperX · PyTorch · RAG · Speech Recognition · Pronunciation Assessment · NLP

PROFESSIONAL EXPERIENCE

LII Lab Sylhet, Bangladesh
Senior Software Engineer 01/2025 – 04/2026

OnePTE & OneIELTS — AI-powered English test-prep SaaS (Web + iOS + Android).

- Architected **multi-tenant SaaS backend** with django-multitenant — full data isolation, custom-domain routing, per-tenant scoring config and feature flags; scaled to two production products on a shared codebase with zero cross-tenant data leakage.
- Designed a **QTI 3.0-compliant exam engine** supporting 100+ question types across IELTS Academic & General with XML parsing, interaction routing, and strategy-pattern band-score normalization.
- Built a **real-time AI speech-evaluation pipeline** (Speech Recognition + Pronunciation Assessment + NLP) delivering end-to-end scoring in under 15 seconds — combining WhisperX, OpenAI APIs, and custom NLP graders for fluency, lexical resource, and grammar.
- Engineered a **multi-gateway payment platform** across 5 providers (Stripe, Razorpay, SSLCommerz, Google Play, Apple) with subscription lifecycle management, webhook idempotency, and transaction-level reconciliation.
- Stood up a production **observability stack** with Prometheus + Grafana — health checks, per-API latency histograms, and error-rate alerting.
- Drove **AI-native engineering practice** across the team — Claude Code workflows, CLAUDE.md skill/subagent design, and AI-assisted PR review; mentored junior engineers through code reviews and architecture walkthroughs.

Stack: Django · DRF · FastAPI · Python · PostgreSQL · Redis · Celery · Docker · GCP · Prometheus · Grafana · Next.js · TypeScript

OnePTE — AI-scored PTE test-prep platform (Flutter + Django).

- **Full-stack ownership from day one** — designed and shipped the Flutter cross-platform app (iOS + Android) covering all four PTE modules and 20+ task types with audio capture, timer management, and offline-tolerant state via Riverpod.
- Built a **task-group mock-test engine** with modular exam templates, configurable time allocations, automated question progression, and multi-dimensional score breakdowns.
- Engineered the **AI scoring backend** for spoken and written PTE tasks using Speech Recognition, Pronunciation Assessment, and NLP for multi-trait evaluation.
- Integrated **subscription billing across 4 platforms** (Stripe, SSLCommerz, Google Play, Apple) with webhook handling, transaction deduplication, and in-app purchase receipt verification.
- Led **90+ production releases** across Android, iOS, and Web — owned release pipelines, Firebase Remote Config, and CI/CD across all channels.

Stack: Flutter · Dart · Riverpod · Django · DRF · Python · PostgreSQL · Redis · Celery · Firebase · Next.js · TypeScript

SELECTED PROJECTS

Sushi Lab — Bilingual Restaurant Ordering & Marketing Platform

2026

Japanese restaurant in Chartres, France — solo end-to-end build.

- Designed and shipped a production **bilingual (FR/EN)** ordering + marketing site on Next.js 16 App Router with React 19 and server actions — translated URL segments and locale-aware checkout.
- Built an **ordering engine** with a hydration-safe Zustand cart, scheduled-ahead time slots, per-item option pickers, and server-enforced time-of-day availability windows.
- Engineered **triple-channel notifications** (Telegram bot, transactional customer email, restaurant inbox) dispatched in parallel, plus AI-discoverable SEO (JSON-LD, per-locale canonical + hreflang) and a GA4 ecommerce funnel.

Stack: Next.js 16 · React 19 · TypeScript · Tailwind · Zustand · Zod · React Email · next-intl · ZeptoMail · Telegram Bot API · GA4

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

Solved **2,000+ problems** across online judges; participated in **300+ contests** over 4 years.

Runner-up (Bangladesh), IEEEExtreme 16.0 (2022) — global rank 149 / 6,500+ teams.

Codeforces (Max: **1603**) · CodeChef (Max: **1965**)

ICPC Asia Dhaka Regional 2020 — finalist (51st); Preliminary 2021 — 87th of 1,700+ teams.

Champion, LU CSE Carnival National Hackathon 2023; **Champion**, LU TechStorm 4 (2021).